

PFRDSFELD, Germany

WMO: 106260

Lat: 49.850N

Lon: 7.600E

Elev: 1299

StdP: 14.02

Time Zone: 1.00 (EUC)

Period: 86-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.7	17.7	8.4	8.9	17.2	12.1	10.7	20.0	29.8	42.5	26.0	39.2	7.5	70	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	82.7	65.1	79.8	64.1	76.9	62.7	67.5	78.9	65.6	76.5	63.9	74.1	8.6	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	90.3	72.1	61.2	84.7	70.0	59.7	80.3	68.4	32.7	78.5	31.2	76.5	29.8	74.4	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.7	21.4	18.7	DB	9.6	89.5	5.5	4.9	5.7	93.0	2.5	95.9	-0.6	98.6	-4.6	102.2
			WB	9.3	70.5	5.6	1.8	5.3	71.8	2.0	72.9	-1.1	73.9	-5.2	75.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	32.7	32.5	39.8	46.0	54.4	58.9	64.2	63.5	55.8	48.2	39.0	35.0	(6)
	DBStd	13.17	8.02	8.14	7.18	7.22	6.69	6.67	6.36	6.60	5.17	6.43	6.36	7.19	(7)
	HDD50	2469	537	489	324	162	34	5	0	0	7	113	331	467	(8)
	HDD65	6517	1002	909	782	569	333	205	96	107	281	521	781	931	(9)
	CDD50	1592	0	0	7	44	169	273	440	420	180	57	1	1	(10)
	CDD65	162	0	0	0	0	3	22	71	62	4	0	0	0	(11)
	CDH74	1309	0	0	0	1	27	172	581	502	26	0	0	0	(12)
	CDH80	269	0	0	0	0	1	12	129	126	1	0	0	0	(13)
(14) Wind	WSAvg	7.6	9.4	8.0	9.1	7.4	6.9	6.7	6.7	6.3	6.9	7.1	7.4	8.9	(14)
(15) Precipitation	PrecAvg	24.8	2.0	1.8	1.7	1.5	2.2	2.3	2.6	2.4	1.9	2.1	2.1	2.3	(15)
	PrecMax	34.8	6.5	4.8	4.9	3.3	4.9	5.8	7.2	4.6	4.4	6.1	3.8	6.5	(16)
	PrecMin	18.0	0.3	0.3	0.2	0.0	0.5	0.5	0.9	0.4	0.6	0.4	0.1	0.3	(17)
	PrecStd	4.0	1.2	1.1	1.1	0.8	1.1	1.2	1.4	1.1	1.0	1.1	0.9	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.5	55.5	64.1	72.7	77.1	82.0	87.9	89.2	78.4	68.9	54.0	53.8	(19)
		MCWB	46.9	48.1	52.8	55.6	60.0	65.1	67.4	66.5	61.8	60.0	50.4	51.1	(20)
	2%	DB	48.1	50.2	57.6	67.6	73.6	78.8	83.4	82.8	71.9	63.3	52.0	50.6	(21)
		MCWB	46.1	45.9	48.6	53.7	58.2	63.9	65.6	65.0	61.1	56.5	49.4	48.3	(22)
	5%	DB	46.1	46.5	53.4	62.7	70.2	75.5	80.4	79.2	68.1	59.5	49.9	48.2	(23)
		MCWB	44.4	44.1	47.1	51.4	57.0	61.9	64.4	64.1	59.6	55.2	47.4	46.7	(24)
	10%	DB	43.3	43.0	50.0	58.5	67.7	71.7	77.0	75.5	64.7	57.1	47.9	45.2	(25)
		MCWB	41.8	41.0	45.1	49.5	56.3	60.4	62.9	62.4	57.2	53.9	46.0	43.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.2	49.3	53.9	57.5	62.9	68.8	69.7	68.6	66.2	61.3	51.8	51.8	(27)
		MCDB	49.8	53.8	62.9	69.1	71.6	78.1	82.5	82.2	72.7	66.7	53.1	53.3	(28)
	2%	WB	46.1	46.6	50.1	54.7	59.9	65.8	67.7	66.7	62.7	57.9	49.8	49.1	(29)
		MCDB	47.5	49.7	54.9	65.6	70.2	74.0	79.5	79.6	69.2	61.4	52.0	50.6	(30)
	5%	WB	43.9	43.9	48.0	52.2	58.2	63.4	65.8	65.1	60.5	56.0	48.1	46.5	(31)
		MCDB	45.2	46.3	51.9	61.4	68.3	73.0	77.1	76.7	66.5	59.1	49.6	48.1	(32)
	10%	WB	41.7	41.1	45.6	49.7	56.6	61.1	63.9	63.4	58.4	54.2	45.9	43.7	(33)
		MCDB	43.2	42.7	49.6	57.1	65.9	69.9	74.6	73.8	63.6	56.9	47.3	45.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	10.1	13.3	15.9	17.2	16.4	17.8	18.3	14.8	12.3	9.3	7.8	(35)
		MCDBR	11.2	14.6	18.7	21.3	21.6	21.8	22.5	23.0	19.8	15.8	11.4	10.0	(36)
	5% WB	MCWBR	9.7	10.8	11.6	11.6	11.6	10.8	10.2	10.6	11.4	10.5	9.0	9.0	(37)
		MCWBR	10.5	14.0	16.3	19.7	19.3	19.4	20.4	20.4	17.7	14.5	10.1	9.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.311	0.338	0.370	0.393	0.399	0.402	0.402	0.397	0.371	0.360	0.333	0.309	(40)
		taud	2.410	2.354	2.281	2.262	2.273	2.288	2.279	2.321	2.374	2.391	2.401	2.400	(41)
	Ebn at Noon		233	252	263	269	272	272	270	266	261	241	220	215	(42)
		Edh at Noon	20	27	35	39	40	40	40	36	31	26	20	18	(43)
(44) All-Sky Solar Radiation	RadAvg		252	481	853	1341	1583	1759	1667	1429	1067	617	285	196	(44)
	RadStd		38	89	129	167	161	158	179	144	130	60	33	29	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page